

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of B.Sc. Examination, April, 2019

Course code & Name: BS 405 & Immunology

Date: 29/04/2019

Day: Monday

Time: 1.30 PM to 2.00 PM

Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of a non - programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	You enter a dusty room, feel an itch in your nose, and sneeze. This is an example of the operation of which of the following innate immune mechanism?		01
	(i) the released granular contents of your granulocytes.	(ii) the low pH of the environment.	
	(iii) the physical barrier produced by hairs.	(iv) phagocytosis by macrophages.	
2.	In order for T cells to respond to the antigen for which they are specific, they need to recognize which of the following?		01
	(i) B cells.	(ii) the antigenic epitope displayed by MHC molecules.	
	(iii) immunoglobulin.	(iv) cytokines.	
3.	The formation of memory immune responses is the objective of vaccination. Immunological memory is predominantly the function of which of the following?		01
	(i) the complement system.	(ii) cells bearing pattern-recognition receptor molecules.	
	(iii) molecules comprising the chemical barrier.	(iv) cells of the adaptive immune system.	
4.	Which of the following mechanisms is used by Herpes Simplex Virus to interfere in adaptive immune response?		01
	(i) Blocking transporter protein (TAP).	(ii) Blocking expression of MHC II	
	(iii) Blocking expression of MHC I	(iv) Both 'i' and 'ii'	

5.	Tissue damage caused by wounds or invading pathogenic organisms induces a complex sequence of events collectively known as		01
	(i) Opsonization	(ii) Inflammation	
	(iii) Phagocytosis	(iv) None of the above	
6.	Cytokines are:		01
	(i) Hormone like polypeptides	(ii) Viruses	
	(iii) Bacteria	(iv) Carcinogens	
7.	Which of the following convey the longest lasting immunity to an infectious agent?		01
	(i) Naturally acquired passive immunity	(ii) Artificially acquired passive immunity	
	(iii) Naturally acquired active immunity	(iv) None of the above	
8.	B and T cells are produced by stem cells that are formed in:		01
	(i) Bone marrow	(ii) Spleen	
	(iii) Liver	(iv) Lymph node	
9.	B cells that produce and release large amounts of antibody are called:		01
	(i) Memory cells	(ii) Basophils	
	(iii) Plasma cells	(iv) Killer cells	
10.	BCG is used to protect against which of the following disease?		01
	(i) Tuberculosis	(ii) Rabies	
	(iii) Malaria	(iv) Hepatitis B	
11.	Active immunisation involves_____.		01
	(i) Administration of antigens	(ii) Administration of antibodies	
	(iii) Both 'i' and 'ii'	(iv) None of the above	
12.	Which of the following are pathogenic attributes of <i>Schistosoma mansoni</i> ?		01
	(i) Have M protein	(ii) Mimics some of the host antigens	
	(iii) Blocks JAK-STAT pathway	(iv) None of the above	
13.	Clonal selection occurs when antigen is encountered by:		01
	(i) Basophils	(ii) Mast cells	
	(iii) Neutrophils	(iv) T cells	
14.	Which category of hypersensitivities is IgE-mediated?		01
	(i) Type I	(ii) Type II	
	(iii) Type III	(iv) Type IV	
15.	Which of the following substances will not stimulate an immune response unless they are bound to a larger molecule?		01
	(i) Antigen	(ii) Hapten	
	(iii) Virus	(iv) Antibody	

16.	How protozoan parasite <i>Plasmodium</i> avoids the innate and adaptive immune response of the host?		01
	(i) Keep changing its antigens	(ii) Deactivates complement	
	(iii) Infect T-cells	(iv) Kills B-cells	
17.	The ability of the immune system to recognize self antigens versus nonself antigen is an example of:		01
	(i) Tolerance	(ii) Specific immunity	
	(iii) Cell mediated immunity	(iv) Humoral immunity	
18.	Which of the following is a non-organ-specific (systemic) autoimmune disease:		01
	(i) Myasthenia Gravis.	(ii) Insulin-dependent diabetes mellitus.	
	(iii) Pernicious anemia.	(iv) Systemic lupus erythematosus (SLE).	
19.	Which of the following is true for <i>Corynebacterium diphtheria</i> ?		01
	(i) Causes tuberculosis	(ii) Produces endotoxin	
	(iii) Colonies in nasopharyngeal tract	(iv) None of the above	
20.	O blood group is universal donor because the blood has		01
	(i) Antigen A	(ii) Antigen B	
	(iii) Both antigens A and B	(iv) No antigens	

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IV Semester of B.Sc. Examination, April, 2019

Course code & Name: BS 405 & Immunology

Date: 29/04/2019 Day: Monday Time: 2.01 PM to 4.30 PM Maximum Marks: 50

Instructions:

- 1. Section I and II must be attempted in TWO ANSWER SHEETS.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of a non - programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed (attempt any 5)		20
1.	Write down the names of different immunodiagnostic techniques? Discuss briefly any one of them.	04
2.	What are the differences between passive and active immunity?	04
3.	What is phagocytosis? Describe briefly the process.	04
4.	What are the difference between MHC Class I and II?	04
5.	How do you make monoclonal antibodies? How can monoclonal antibodies be used?	04
6.	Write a short note on Immunosuppression.	04
7.	What is Autoimmunity? What causes an autoimmune disease?	04
8.	What are the key cells of the immune system?	04
SECTION – II		
Q-III Answer the following questions as directed (attempt any 10)		30
1.	What are the contributions of scientists in the field of Immunology?	03
2.	What are the main functions of Immunoglobulin?	03
3.	What are cancers of the immune system? What kind of cancer can be treated with immunotherapy?	03
4.	What are the different types of hypersensitivity? Discuss briefly.	03
5.	What are live attenuated vaccines? Enlist advantages and disadvantages with any two examples?	03

6.	How do vaccines give immunity?	03
7.	Describe any three pathogenic attributes of viruses.	03
8.	What is transplantation? Why transplants are rejected?	03
9.	Enlist the reasons, why host mounts a low or weak immune response to <i>Plasmodium</i> ?	03
10.	What are the complement proteins? What are the role of complement proteins in the immune system?	03
11.	What are the differen phases of adaptive immune responses? Discuss briefly.	03
12.	How is recombinant DNA used to make vaccines? What is the difference between recombinant vaccine and DNA vaccine?	03
13.	Describe how innate immunity plays an important role in preventing fungal infections.	03
14.	What are subunit vaccines? Enlist advantages, disadvantages and any two examples of the subunit vaccines?	03